

Introduction to Data & Study Design

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MATH 240

Lecture #2

Reminders

- Homework #0 is due on Friday, 9/4 at 11:59pm
 - ▶ Includes survey, setup of tech, and turning in quick problem
- Office hours: none on Monday, Tuesday from 3-4:30pm in Rocky 106A

Activity: Statistics

Suppose we want to investigate the average number of hours slept daily by Vassar students.

- Grab a Post-It note and write the number of hours you slept last night and place it on the board.
- Consider the following:
 - ▶ Population of interest?
 - ▶ Sample gathered?
 - ▶ Characteristics of sample?
 - ▶ Variable of interest?
 - ▶ Statistic?
 - ▶ Parameter?

Terms to Consider

- Observation vs. variable
- Types of variables
- Explanatory vs. response
- Observational study vs. experiment

Learning Objectives

- **Identify** parts of a data frame and characteristics of a study/experiment
- **Distinguish** between a population and sample, and connect how a study portrays a population
- **Differentiate** between various types of variables, including nominal, ordinal, discrete, and continuous variables

Demo: Tree Data

For the next 5-10 minutes, we will explore the following tree dataset from NYC Open Data ([Link](#)). We will investigate the following questions:

- ① What do the rows represent? What about the columns?
- ② Pick three columns from the dataset. What values exist in each column? Are they grouped or numeric?
- ③ Click on the 'About' section in the webpage and read the dataset description. What do the data represent? Can these data be generalized to other areas?

The Big Picture

Statistical questions are framed in terms of **populations** and **parameters**. However, to answer these questions, we often focus on a **sample** and a **statistic**.

Population	Sample
All likely U.S. voters	2000 individuals in political poll
All Vassar students	100 random students polled out in the open
Flies in New York	1000 collected flies from multiple areas

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- 1 What does our data look like?
- 2 Where does our data come from?

Parts of the Data

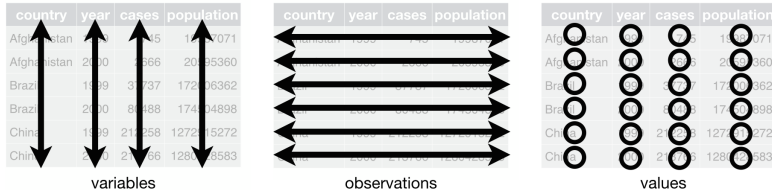


Figure: Diagram of Parts of Data Frame (Garrett & Wickham, 2019)

- **Variable:** characteristic being measured
 - ▶ Usually represented by a single **column** in dataset
 - ▶ e.g. preferred political candidate, class level, health status

Parts of the Data

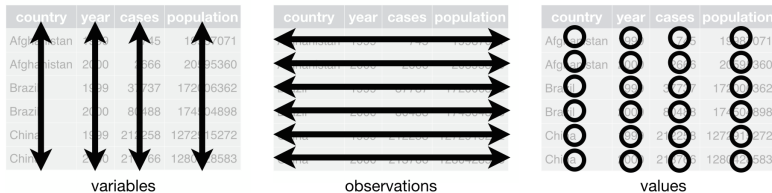


Figure: Diagram of Parts of Data Frame (Garrett & Wickham, 2019)

- **Observational unit:** person or thing on which measurements are taken on
 - ▶ i.e. an observation
 - ▶ Usually represented by a single **row** in dataset
 - ▶ e.g. one U.S. voter, one Vassar student, one fly in New York

Parts of the Data

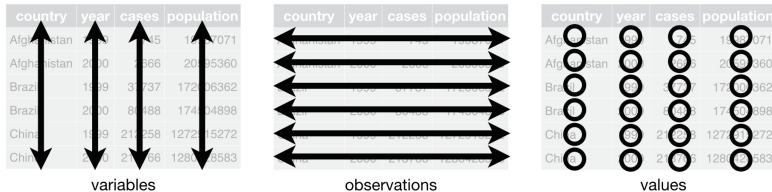


Figure: Diagram of Parts of Data Frame (Garrett & Wickham, 2019)

- **Data frame:** collection of data, where a row is an observational unit and a column is a variable
 - ▶ Each cell is a single value (referred to as **tidy data**)
 - ▶ e.g. spreadsheet of responses

Variable Types

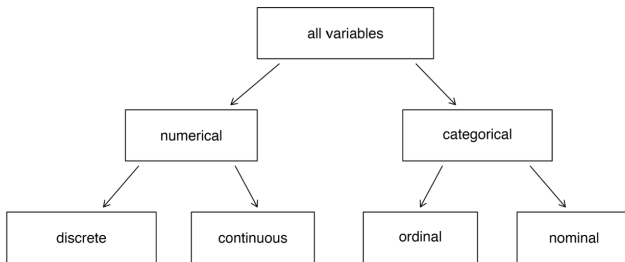


Figure: Breakdown of Types of Variables (IMS, Ch. 1)

Numeric Variables

tree_dbh

3
21
3
10
21
11
11
9
6
21
11
8
13
22
30

- **Discrete:** numerical values with jumps

- ▶ ←

- ▶ Typically count data (e.g. 1, 2, 3, ...)

- **Continuous:** numerical variable that can have infinite values within range; no jumps

- ▶ →

- ▶ e.g. those represented as decimals usually

longitude

-73.8442
-73.8187
-73.9366
-73.9345
-73.976
-73.9849
-73.9853
-73.9873
-74.0763
-73.9697
-73.9112
-73.912
-74.1627
-73.9682
-73.9211

Categorical Variables

sidewalk

NoDamage

Damage

Damage

Damage

Damage

NoDamage

NoDamage

NoDamage

NoDamage

NoDamage

NoDamage

NoDamage

NoDamage

NoDamage

Damage

- **Nominal:** categorical variable whose ordering does not matter

- ▶ ←

- ▶ e.g. names, eye colors, regions

- **Ordinal:** categorical variable whose ordering does matter

- ▶ →

- ▶ e.g. class level, Olympic medals,

health

Fair

Fair

Good

Good

Good

Good

Good

Good

Good

Fair

Good

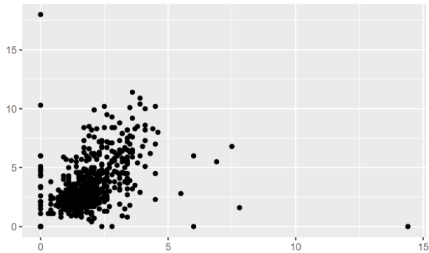
Poor

Fair

Good

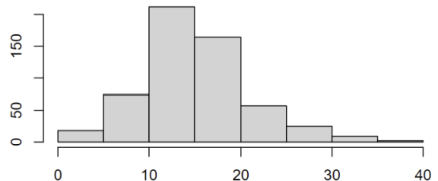
Fair

Seeing Data



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	11.30	14.40	15.09	18.30	37.60

BRK	CHI	CHO	CLE	DAL
17	12	15	17	17
SAC	SAS	TOR	UTA	WAS
17	17	16	16	19



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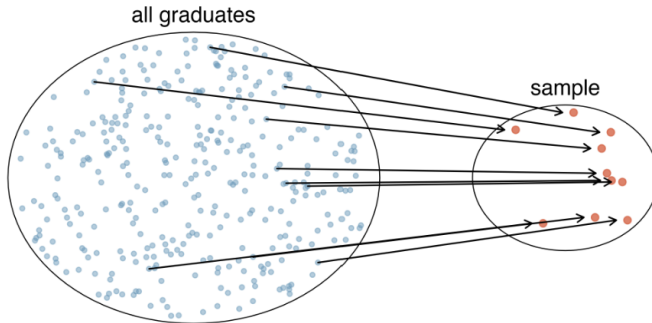
- 1 How do we collect data properly?
- 2 What are some things we need to keep in mind when collecting data?

Activity: Population vs. Sample

Your team is tasked with asking a very important question: how many cups of coffee/matcha does a person in the Vassar community drink on a day-to-day basis? Discuss and write down answers to the following questions:

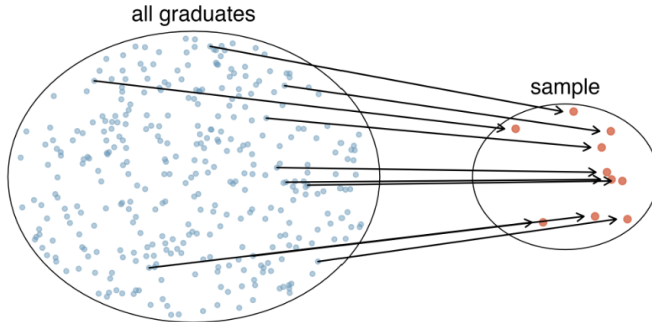
- 1 How would you go about asking this question? Who would you ask?
- 2 What problems might arise when trying to investigate this question?
- 3 What might be the best way to answer the question once you have data from many participants?

Population vs. Sample



- **Population:** target grouping the research question is centered around
 - ▶ If you can get a measurement from everyone/thing → census
 - ▶ Otherwise, you need to take a **sample**

Population vs. Sample



- **Sample:** subset taken from population
 - ▶ Should be **representative** of population

Parameters and Statistics

- **Parameter:** Calculated value representing the population
 - ▶ $\mu, \rho, \mu_1 - \mu_2, \sigma$
- **Statistic:** Calculated value based on sample of data; attempts to estimate parameter
 - ▶ $\bar{x}, \hat{p}, \bar{x}_1 - \bar{x}_2, s$

Sampling from a Population

- **Goal:** an effective and unbiased sample that represents the whole of the population
 - ▶ Tricky to do!
 - ▶ Many factors and variables at play, leading to **bias**
- Seeking **random selection** from a population
- **Example:** Picking 10 customers throughout the day from a Starbucks. Think about...
 - ▶ how do we pick these 10 customers?
 - ▶ how do we contact these 10 customers?
 - ▶ will these 10 customers actually respond?
 - ▶ do they precisely represent the population?

Characteristics of a Sample

- Goal is **random sampling**
 - ▶ Each individual in population has known, non-zero probability of being included in sample; is random
- Watch out for:
 - ▶ Anecdotal evidence
 - ▶ Convenience sample
 - ▶ Biases (non-response, selection, measurement)
 - ★ Systematic error affecting accuracy of a parameter

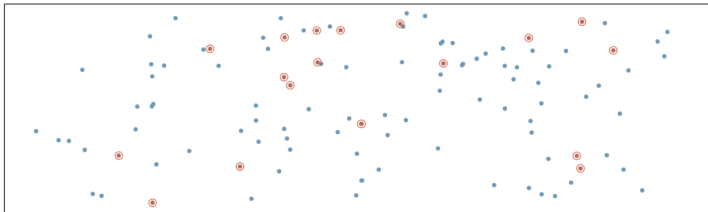
Four Sampling Methods

- ① Simple random sampling (SRS)
 - ▶ Most intuitive; equivalent to picking names out of a hat
- ② Stratified sampling
 - ▶ divide-and-conquer; divide similar cases/makeup (usually with another variable, run SRS)
- ③ Cluster sampling
 - ▶ Break population into heterogeneous clusters; sample all observations from chosen clusters
- ④ Multistage sampling
 - ▶ Break population into clusters randomly; sample randomly within each selected cluster

Examples of the Four Sampling Methods

Let's return to the coffee research problem mentioned at the beginning. Recall that the target population is everyone in the Vassar community.

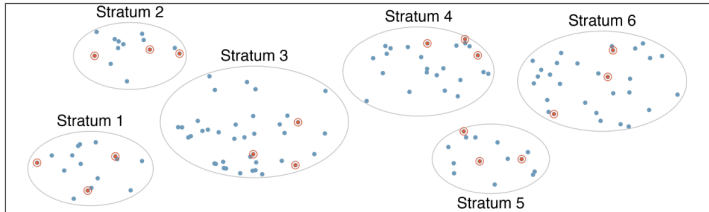
- 1 **Simple random sampling (SRS):** with a list of all community members' names, we pick 200 names randomly from the list



Examples of the Four Sampling Methods

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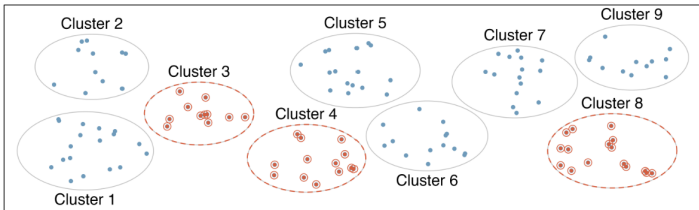
- 2 **Stratified sampling:** Isolating members into strata representing their status at the college (e.g. students, faculty, employees, community members), we pick 50 members randomly from each of the strata



Examples of the Four Sampling Methods

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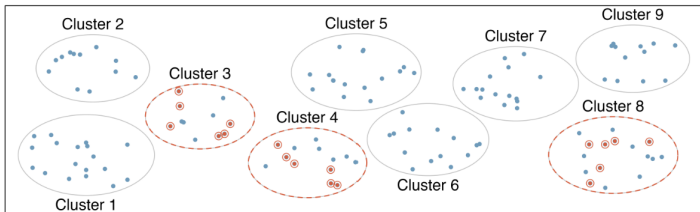
- ③ **Cluster sampling:** break members randomly into many groups of 40, pick 5 clusters and select all members in those chosen clusters



Examples of the Four Sampling Methods

Let's return to the coffee research problem mentioned at the beginning. Recall that the target population is everyone in the Vassar community.

- 4 **Multistage sampling:** break members randomly into many groups of 40 students, pick 10 clusters and select 20 members randomly in those chosen clusters



Experiments vs. Observational Studies

- In lab settings, researchers may want to assign treatments to cases
 - ▶ This type of assignment is called an **experiment**
 - ▶ Used to show causal relationship between two variables
- In cases where we use no treatment, we are dealing with **observational data**
 - ▶ Used to show associations or form hypotheses

See You Friday!

- Bring laptops for Friday's class! We will be working on the first lab.
- Read through chapter 3
- Homework #0 (due Friday, 9/4)
 - ▶ Introductory assignment consisting of setup, survey, and brief problem